FREELANCE · SOFTWARE ENGINEERING

Jens Laufer

Senior product engineer for business software — the agents write the code, I own the requirements and verify what they deliver

Requirements & specs for coding agents · acceptance criteria & independent tests · quality gates & drift control · business software (invoicing, billing, booking, master data) · modernisation to go-live · Java / Spring Boot · Python · Vue / TypeScript · PostgreSQL / Oracle / MySQL · Docker / Kubernetes · CI/CD

Since 10/2025 I work the way this project describes: coding agents write the code, and my job is to break the work down into specs with acceptance criteria, then verify what comes back against the spec and the diff, not against the agent's own report. I build and run that in production — an autonomous AI business partner, Fabrik, where worker agents turn tickets into pull requests in parallel, an unattended coding pipeline with 759 tasks to date, and an agent that runs the finance back office of a GmbH with a human approving every filing. Before that, 29 years of business software inside client teams: the full invoice life cycle at the Deutsche Verrechnungsstelle, heating cost billing at Brunata, the re-build of a booking and reservation system at DER, master data migration at Commerzbank, and a Java EE to Spring Boot migration at Condor.

AVAILABLE	WORKS	REMOTE SHARE	BASED IN
immediately	**remote only**	**100 %** 0 % on-site	**Karlstein am Main** Germany

FOCUS

Coding agents in daily production use since 10/2025: the agent writes the code; I write the task — repository, scope, acceptance criteria — and review the pull request. Merging stays with a human.

Verification with independent tests: a quality gate (unit tests, build, review) must pass before any push; a test must reproduce a bug red before the fix may turn it green; more than 40 checks test an agent's work against the world instead of its own report.

Business software end to end: full invoice life cycle (creation, payment, dunning, debt collection), heating cost billing in an industry file format, booking and reservation services, invoice handover to **SAP** over an ESB.

Modernisation to go-live: Java EE payment back-end migrated to Spring Boot (Condor); master data migration at a major bank (Commerzbank); a running prototype taken over and put live at Deutsche Telekom.

Freelance since 1997, embedded in the client's team — from discovery to production support and handover, 100 % remote.

ROLES

Full-stack engineering · Solution design · Backend engineering · DevOps · Build & configuration management · Data analysis · Prototyping · Machine learning · Jenkins plugin development · Maven plugin development

SKILLS

LANGUAGES

Java · JavaScript · TypeScript · Python · R · Groovy

MICROSERVICES / BACKEND

Spring Boot · Spring · Spring Data · Spring Integration · Spring Security · OAuth2 · Microservices · Feign · Sleuth · Zipkin · Python Eve · Python Flask · Java EE · EJB

PERSISTENCE / DATABASES

MySQL · MongoDB · PostgreSQL · Oracle · Neo4j · JPA · Hibernate

API / INTEGRATION

REST · Swagger · OpenAPI · SOAP · Webservices · CXF · XFire · Kafka · ServiceMix

FRONTEND

Vue.js · React · AngularJS · GWT · Quasar · HTML · CSS · Bootstrap

DEVOPS / CLOUD

Docker · Kubernetes · Rancher · GitLab CI · Jenkins · Maven · Grunt · npm · Artifactory · AWS (EC2, S3, SageMaker, VPC) · Microsoft Azure · DigitalOcean

SECURITY / IAM

Keycloak · Spring Security · OAuth2

TESTING

JUnit · TDD · Mockito · Selenium · DBUnit · Playwright

MACHINE LEARNING

scikit-learn · Keras · DeepLearning4J · CNN · Transfer Learning · Supervised/Unsupervised · k-means · PCA · t-SNE · Reinforcement Learning · OpenAI Gym · Topic Modeling

DATA SCIENCE

pandas · numpy · dplyr · Jupyter · RMarkdown · ggplot · seaborn · Vega-Lite · A/B Testing · Webscraping

TOOLING / OTHER

Git · GitLab · Subversion · Jira · Confluence · Kibana · ANTLR · JavaCC · ZXing (QR) · PayPal API · RDF/XML/JSON · SAP (BAPI) · Phonegap · JBoss · Tomcat

PROJECT HISTORY (SELECTION FROM 2009)

since 03/2026
ongoing

Autonomous coding pipeline — agents work a task queue unattended

agent-tasks · Solytics GmbH · Remote — Software / AI

Harness engineering · Agent engineering · DevOps

Queue-driven harness in which coding agents work through tasks with nobody at the keyboard. A task file names repository, scope, acceptance criteria and model; a runner on own hardware executes it in an isolated worktree, runs the **quality gate** (unit tests, build, review) before any push and opens a pull request — merging stays with a human. Model chosen per task against a 5-hour token budget, hard timeouts per run, at most one new pull request per repository per night. **759 tasks** written to date across the own product repositories.

Claude / Anthropic API · Python · Docker · TDD · systemd

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since 04/2026
ongoing

Fabrik — agent harness for parallel software delivery

Fabrik (fabrikhq.com) · Solytics GmbH · Remote — Software / AI

Harness engineering · Architecture · Full-stack engineering

A harness in which worker agents turn tickets into pull requests in parallel (fabrikhq.com). A **control plane** (FastAPI, Vue) spawns one agent container per repository through the Docker socket. Each agent runs a fixed loop — ticket sync (GitHub, GitLab, Jira), an actionability review that rejects vague tickets, a plan, test-first implementation, a **quality gate** in a clean sidecar container, then the PR; it never merges on its own. Energy budget per agent, feedback from PR comments back into the loop, one VPS per customer behind Caddy.

Claude / Anthropic API · Multi-Agent-Orchestration · Python · FastAPI · Vue 3 · Docker · Caddy · GitHub Actions · SQLite · TDD

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since 10/2025
ongoing

Autonomous AI business partner — Harness & agentic engineering

Solytics GmbH · Remote — AI / Automation

Harness engineering · Agent engineering · AI system architecture · DevOps

A self-directing agent built on Claude that plans, writes code, tests, deploys and reports around the clock. Built as a **harness engineer** — the engineering around the model: the **agent loop** and its stop conditions, **tool use over MCP**, layered **memory** (durable facts, running context, session state), **approval gates** for anything irreversible, **observability** over every run, **budget and rate-limit control** with model routing, plus recovery when a run is killed mid-flight. More than 40 purpose-built checks catch the agent's mistakes before they reach a human. Event-driven over Telegram, GitHub and systemd; runs on own hardware behind a local multi-model proxy.

Claude / Anthropic API · LLM-Agenten · Agent Harness · Tool Use / MCP · RAG · Prompt Engineering · Python · systemd · Telegram Bot API · Docker · TDD

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since 03/2026
ongoing

Verification and drift controls for LLM agents

Solytics GmbH · Remote — AI / Automation

Evaluation & verification · Reliability engineering

More than 40 purpose-built checks that test an agent's work against the world instead of against its own report: every link in an outgoing message is resolved before it is sent, promises the agent made are tracked until delivered, messages without an answer are flagged, pull-request mergeability is computed with git instead of read from a cached API value, and a test run that never started fails instead of passing silently. Each check fails closed — unknown is never reported as healthy — and has its own tests. Watchdogs over the agent's services and credentials, with a second alert path for when the primary channel is the one that broke.

Claude / Anthropic API · Python · Docker · TDD

since 03/2026
ongoing

LLM agent in the finance back office of a GmbH

Solytics GmbH · Remote — Finance / Compliance

Agent integration · Domain design · Back-end engineering

An agent that runs the bookkeeping of a German GmbH end to end, with a human approving every filing. It files bank statements and receipts, books double-entry on SKR04 through a purpose-built CLI, reconciles every account against the statement, and checks VAT and payroll returns against the journal before they are submitted — once catching a field that would have lost the input tax. Tools beneath it: an API for **XRechnung 3.0** and **ZUGFeRD 2.3** with five validation layers (Pydantic, BR-DE rules, arithmetic, XSD, Schematron), balance sheet, P&L, E-Bilanz and **DATEV export**.

Python · Pydantic · XRechnung / ZUGFeRD (EN 16931) · Polars · FastAPI · Docker · TDD

07/2022–04/2024
22 months

Front- and back-end, heating cost billing

Brunata · Remote — Industry

Full-stack engineering · Solution design · DevOps

Production server-side services and APIs in Java and Spring Boot: a microservice that parses and generates heating-cost files in the Haweiko-Arge industry format. Designed and built in a microservices architecture with REST interfaces (OpenAPI, Swagger), secured with Spring Security and Keycloak, persisted in MongoDB. Built and maintained the GitLab CI pipelines for automated build and deployment. Worked to DevOps principles, owning build and operation of the services in Docker. Test-driven development (JUnit, Mockito), automated end-to-end tests with Playwright. Front-end development.

Java · Spring Boot · MongoDB · JUnit · TDD · React · Docker · GitLab · GitLab CI · Spring Security · Keycloak · Mockito · OpenAPI · Swagger · REST · Microservices · Playwright

01/2022–06/2022
6 months

Payment back-end migration Java EE → Spring Boot

Condor · Remote — Marketing

Backend engineering

Migrated an EJB-based Java EE payment back-end to Spring Boot and shipped it as a Docker container.

Java · Java EE · JPA · Spring Boot · MySQL · Docker · Mockito · JUnit · TDD

12/2019–12/2020
13 months

Web-based configuration of telecommunications devices

Deutsche Telekom · Darmstadt/Remote — Telecommunications

DevOps · Build & configuration management · Full-stack engineering

Took over an existing prototype for configuring telecommunications devices over a web interface, developed it further and moved it into production. Production server-side services and APIs in Java and Spring Boot in a microservices architecture (REST, Feign, Spring Data/JPA), Vue.js front-end, secured with Keycloak, persisted in MySQL and MongoDB. Worked to DevOps principles, owning build, operation and monitoring of the services: Docker and Kubernetes (Rancher), logging and tracing via Kibana, Sleuth and Zipkin; built and maintained the GitLab CI pipelines.

Java · Spring Boot · Docker · Kubernetes · Microservices · REST · Spring Data · JPA · JUnit · TDD · Mockito · Vue.js · MySQL · MongoDB · HTML · CSS · GitLab · GitLab CI · Rancher · Kibana · Feign · Keycloak · Sleuth · Zipkin

07/2015–07/2017
25 months

Invoice management platform (SME)

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Platform covering the full invoice life cycle (creation, payment, dunning, debt collection); a range of Spring Boot microservices.

Spring Boot · Docker · Kubernetes · Microservices · REST · Java · Maven · Spring Security · OAuth2 · Spring Integration · MongoDB · Jira · Confluence · JUnit · TDD · Mockito · Selenium · AngularJS · Grunt · npm · Spring · Git

02/2015–06/2015
5 months

Invoice management platform (SME) — debt-collection integration

Deutsche Verrechnungstelle / DVAG · Frankfurt a. Main — Insurance, Finance, Startup

Full-stack engineering

Microservice integrating a debt-collection provider, including a single-page admin front-end.

Spring Boot · Docker · Kubernetes · Microservices · REST · Java · Maven · Spring Security · OAuth2 · Spring Integration · MongoDB · Jira · Confluence · JUnit · TDD · Mockito · Selenium · AngularJS · Grunt · npm · Spring · Git

06/2012–12/2013
19 months

Business services for a booking and reservation system

Deutsches Reisebüro (DER) · Frankfurt a. Main — Travel

Backend engineering

Implementation of business services as part of the re-build of a booking and reservation system.

Java · JUnit · TDD · Mockito · Spring · SOAP · Webservices · Jira · Maven

04/2010–09/2010
6 months

Web services on an ESB (mediation and rating)

1&1 · Montabaur — Telecommunications

Backend engineering

Software design, web services, maintenance of the rating engine; web services that hand invoices over to **SAP**.

Java · Spring · Spring Security · JPA · Hibernate · Maven · JUnit · TDD · Mockito · DBUnit · ServiceMix · Webservices · CXF · XFire · SAP-Anbindung · SAP BAPI · JBoss · Oracle · Subversion (SVN)

05/2009–03/2010
11 months

Master data migration at a major bank

Commerzbank · Frankfurt a. Main — Banking, Finance

Backend engineering

Workflows for persisting master data, a canonical XML model, RDF description for machine processing.

Java · Java EE · IBM MDM (WCC) · EJB · JPA · RDF · XML · Oracle · Subversion (SVN)

CONTACT & CREDENTIALS

CONTACT & PROFILES

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BASED IN Karlstein am Main, Germany

EDUCATION

11/1997

Dipl.-Ing. (Univ.) Electrical Engineering (MSc equivalent)

University of Erlangen-Nuremberg

WEBSITE	jenslaufer.com
GITHUB	github.com/jenslaufer
LINKEDIN	linkedin.com/in/jenslaufer
KAGGLE	kaggle.com/jenslaufer
PUBLICATION	Towards Data Science — author

05/1989

Abitur (university entrance qualification)

Gymnasium am Romäusring, Villingen-Schwenningen

CERTIFICATES

12/2018

Machine Learning Engineer Nanodegree

Udacity

Verify →

08/2017

Data Analyst Nanodegree

Udacity

Verify →